

Lesson Demo · one supervised session · **adult demonstrator owns every flame and gas control**

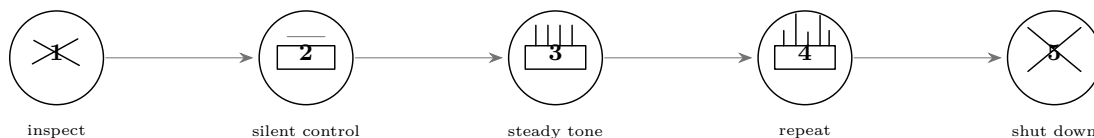
Demo: Rubens' Tube

See, measure, and verify a standing wave safely

Observe in order — do not change two things at once

Lesson Sequence · predict, control, observe, repeat · operator calls every hold point

A controlled observation sequence

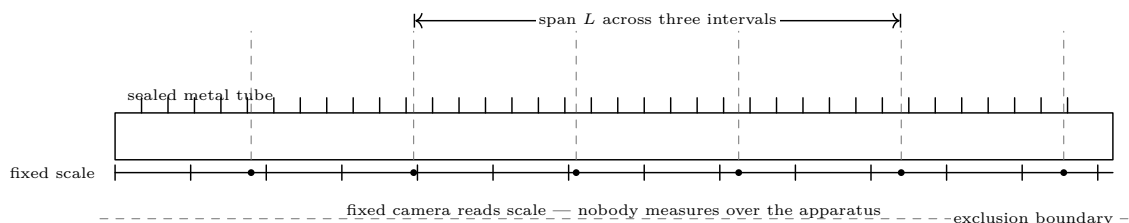


1. **Inspect.** The operator completes the maker's cold checklist. *Check:* apparatus, supports, barrier, weather, extinguisher, and shutoff all pass. *Fail:* tag out; no fuel is admitted.
2. **Observe the silent control.** Students watch from the assigned position. *Check:* the operator calls the row stable and the recorder logs its appearance. *Fail:* operator shuts down; reject the run.
3. **Apply one steady tone.** The operator uses one documented permitted setting. *Check:* the pattern settles and observers mark centres from the fixed scale. *Fail:* operator shuts down; record no spacing.
4. **Repeat the same setting.** The operator alone restarts according to the maker's instructions. *Check:* centres recur within the stated reading uncertainty. *Fail:* retain both records and reject the claim.
5. **Shut down.** The operator follows the maker's shutdown procedure. *Check:* fuel is isolated and the apparatus is cool before the area is released. *Fail:* keep the boundary closed and follow the emergency plan.

Fault gate — stop means stop

If the pre-use checklist fails, the row is irregular before sound is applied, the pattern changes without a commanded change, flame appears anywhere except the intended row, wind disturbs the demonstration, a support moves, the speaker seal changes, or any person crosses the boundary: the run fails. The operator shuts down if safe, isolates the apparatus, and has the defective part remade or professionally repaired. Never patch, tighten, drill, or adjust a fueled or hot apparatus. A failed run produces no measurement.

Measure node spacing from a safe distance



Calculation

Choose several consecutive repeated short regions. Mark their centres on the retained image. Measure L from first to last. Count the N intervals, not marks:

$$d = \frac{L}{N}, \quad \lambda \approx 2d, \quad v = f\lambda.$$

The last relation is a model estimate. Label it with the gas, apparatus, tone, date, and uncertainty; do not report it as a universal constant.

Measurement questions

1. Why is a span across several intervals better than one gap?
2. Did repeats place each short region at the same scale mark?
3. Which centre was hardest to locate? How much could it move?
4. Does the silent control contain an equally regular spacing?
5. Would your conclusion survive the largest reasonable marking error?

Demo: Rubens' Tube

Record observations before doing arithmetic

See, measure, and verify a standing wave safely

Lesson Worksheets · controls, repeats, and traceable marks · retain rejected runs as evidence

Observation record

Run	Tone / ID	Short-region positions from fixed scale	Stable?	Keep / reject
control	silent			
1A				
1B repeat				
2A				
2B repeat				

Measurement record

Run	f / source	first–last	L	intervals N	$d = L/N$	uncertainty / note
repeat						

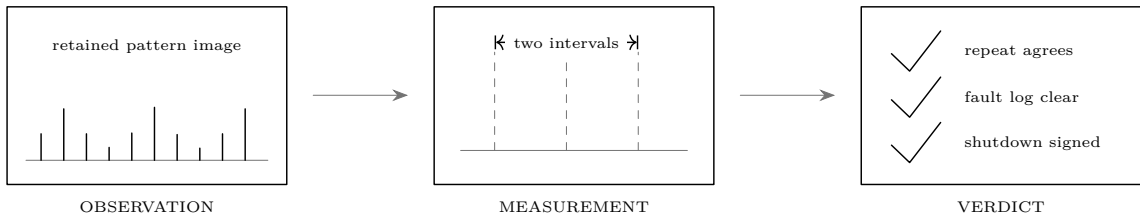
Accept The cold inspection passed; control was recorded; the pattern remained stable; repeated marks agree within stated reading uncertainty; and every number is traceable to a retained run.

Reject Any safety fault, missing control, moving pattern, ambiguous centres, undocumented setting, or disagreement larger than the stated uncertainty. Cross out the arithmetic but keep the row as evidence of why it failed.

Remake decision A damaged, leaking, distorted, or unstable component is not “good enough for one more run.” Tag the apparatus out. A qualified adult decides whether the maker replaces the component or the entire assembly.

Lesson Final proof · inspection, repeatability, shutdown · a photograph alone is not proof

Final verification plate



Inspection and disposition record

Hold point	Evidence / record ID	Operator initials
Cold apparatus checklist passed		
Area, barrier, weather, extinguisher, shutoff passed		
Silent control retained		
Steady-tone run and same-setting repeat retained		
Node centres and uncertainty recorded		
Faults resolved by stop / repair / remake before any later run		
Fuel isolated; apparatus cooled; area released		

Final claim

Complete only after shutdown

At the documented tone, the retained observations ☐ do ☐ do not show fixed short regions that recur at the same positions. The silent control ☐ does ☐ does not show the same regular pattern. Repeated spacing estimates agree within $\pm$ _____. Therefore the evidence ☐ supports ☐ does not yet support the standing-wave model for this session.

Observer: _____ Date: _____ Adult operator: _____

Proof: passed inspection, explicit control, two agreeing observations, traceable measurements, a clear fault record, and confirmed shutdown.
Not proof: a dramatic flame, one image, selected “best” gaps, or a run after a stop condition. A fixed, repeatable pattern tied to a steady tone supports stationary pressure regions; the record makes that conclusion defensible.